

Thirupathi Burra

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PROFILE SUMMARY

AI/ML Engineer specializing in **Generative AI, RAG, LLM integration, and multi-agent systems** with production-grade experience in fintech and compliance AI. Published IJESAT researcher; expert in **LangChain, CrewAI, FastAPI, Docker, and vector databases**. Finalist – Google Agentathon & Smart India Hackathon; Top Learner – Google Cloud GenAI Exchange Program.

EDUCATION

Matrusri Engineering College

Hyderabad, India

B.E. in Computer Engineering

CGPA: 8.00 / 10 (2022 – 2026)

WORK EXPERIENCE

AI/ML Engineer Intern — Infosys Springboard

Jun 2025 – Aug 2025

- Architected an **end-to-end KYC verification and fraud detection pipeline** using OCR, feature engineering, and ensemble ML classifiers, **improving automated accuracy by ~35%** and cutting manual review time.
- Engineered **real-time Flask REST APIs** for low-latency model inference with sub-second response times on concurrent document verification requests.
- Containerized the inference service with **Docker**, reducing deployment time by ~40%; implemented **model monitoring pipelines** ensuring >99% uptime on financial data.

PROJECTS

FinSmartAI — AI Personal Finance & Investment Assistant

GitHub

Python, FastAPI, LLMs, RAG, AstraDB, LangChain, NLP, React

- Designed a **full-stack GenAI fintech platform** for document Q&A, expense intelligence, market sentiment analysis, and stock insights with **citation-backed responses**.
- Built a **RAG pipeline with AstraDB vector search** achieving **BERTScore F1 0.8605** and **Semantic Similarity 0.834**, significantly reducing LLM hallucinations in financial queries.
- Published peer-reviewed research in **IJESAT (Vol. 26, Issue 4, Apr 2026)** – Paper PDF | Journal Page

Audit IQ — Autonomous AI Compliance Auditor

GitHub

Python, FastAPI, LangChain, CrewAI, Multi-Agent AI, RAG, Vector DBs

- Developed a **multi-agent AI system** using CrewAI for automated compliance validation, anomaly detection, and dynamic risk scoring; **reduced audit time by ~60%**.
- Implemented **RAG-based regulatory reasoning workflows** for explainable, citation-backed audit decisions with traceable decision chains.

AI-Powered Identity Verification & Fraud Detection

GitHub

Python, OCR, Computer Vision, ML, Flask, Docker

- Built a **multimodal OCR + ML pipeline** for identity document verification and fraud anomaly detection; containerized with **Docker** and integrated **real-time monitoring dashboards** for API audit traceability.

TECHNICAL SKILLS

Languages: Python, SQL

Generative AI & LLMs: LLM Integration, Prompt Engineering, LangChain, CrewAI, RAG, Agentic AI, Semantic Search, Ollama, Hugging Face Transformers, FAISS, Pinecone, ChromaDB, AstraDB

ML & Deep Learning: TensorFlow, PyTorch, Scikit-learn, NLP, Computer Vision, OpenCV, Model Evaluation

Data Science: Pandas, NumPy, EDA, Feature Engineering, Statistical Modeling, ETL Pipelines

Backend & MLOps: FastAPI, Flask, REST APIs, Docker, Git, Streamlit, Model Deployment, Model Monitoring

Databases: MySQL, MongoDB **Core:** DSA, OOP, DBMS, System Design

ACHIEVEMENTS

- National Finalist — Google Agentathon:** Engineered production-grade multi-agent AI systems competing at national level.
- National Finalist — Smart India Hackathon (SIH):** Built scalable, impact-driven software selected among top national teams.
- Top Learner — Google Cloud GenAI Exchange Program:** Ranked among top performers in Generative AI concepts and implementation.
- Certifications:** Google Cloud AI/ML & GenAI Skill Badges | Infosys Springboard AI/ML Program